MPG No. 1.1.1 1 [Type I, triclinic] [T tensor]

* Rank 0 tensor.

$$[T]$$

$$T = \mathbb{T}_0^{(1)}$$

* Rank 1 tensor.

$$\begin{bmatrix} T_x & T_y & T_z \end{bmatrix}$$

$$T_x = \mathbb{T}_x^{(1)}$$
$$T_y = \mathbb{T}_y^{(1)}$$
$$T_z = \mathbb{T}_z^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} T_{xx} & T_{xy} & T_{xz} \\ T_{xy} & T_{yy} & T_{yz} \\ T_{xz} & T_{yz} & T_{zz} \end{bmatrix}$$

$$T_{xx} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(1)} + \mathbb{T}_v^{(1)}$$
$$T_{xy} = \mathbb{T}_{xy}^{(1)}$$
$$T_{xz} = \mathbb{T}_{xz}^{(1)}$$
$$T_{yy} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(1)} - \mathbb{T}_v^{(1)}$$
$$T_{yz} = \mathbb{T}_{yz}^{(1)}$$
$$T_{zz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_u^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & T_{xy} & T_{xz} \\ -T_{xy} & 0 & T_{yz} \\ -T_{xz} & -T_{yz} & 0 \end{bmatrix}$$

$$T_{xy} = \mathbb{M}_z^{(1)}$$
$$T_{xz} = -\mathbb{M}_y^{(1)}$$
$$T_{yz} = \mathbb{M}_x^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} T_{xxx} & T_{xxy} & T_{xxz} \\ T_{yyx} & T_{yyy} & T_{yyz} \\ T_{zzx} & T_{zzy} & T_{zzz} \\ T_{yzx} & T_{yyz} & T_{zzy} \\ T_{xxz} & T_{yzx} & T_{zzx} \\ T_{xxy} & T_{yyx} & T_{yzx} \end{bmatrix}$$

$$T_{xxx} = 3\mathbb{T}_{x}^{(1)} + 2\mathbb{T}_{x}^{\alpha(1)}$$
$$T_{xxy} = \mathbb{T}_{y}^{(1)} - \mathbb{T}_{y}^{\alpha(1)} - \mathbb{T}_{y}^{\beta(1)}$$
$$T_{xxz} = \mathbb{T}_{z}^{(1)} - \mathbb{T}_{z}^{\alpha(1)} + \mathbb{T}_{z}^{\beta(1)}$$
$$T_{yyx} = \mathbb{T}_{x}^{(1)} - \mathbb{T}_{x}^{\alpha(1)} + \mathbb{T}_{x}^{\beta(1)}$$
$$T_{yyy} = 3\mathbb{T}_{y}^{(1)} + 2\mathbb{T}_{y}^{\alpha(1)}$$
$$T_{yyz} = \mathbb{T}_{z}^{(1)} - \mathbb{T}_{z}^{\alpha(1)} - \mathbb{T}_{z}^{\beta(1)}$$
$$T_{zzx} = \mathbb{T}_{x}^{(1)} - \mathbb{T}_{x}^{\alpha(1)} - \mathbb{T}_{x}^{\beta(1)}$$
$$T_{zzy} = \mathbb{T}_{y}^{(1)} - \mathbb{T}_{y}^{\alpha(1)} + \mathbb{T}_{y}^{\beta(1)}$$
$$T_{zzz} = 3\mathbb{T}_{z}^{(1)} + 2\mathbb{T}_{z}^{\alpha(1)}$$
$$T_{yzx} = \mathbb{T}_{xyz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & T_{xxy} & T_{xxz} \\ T_{yyx} & 0 & T_{yyz} \\ T_{zzx} & T_{zzy} & 0 \\ T_{yzx} & -\frac{T_{yyz}}{2} & -\frac{T_{zzy}}{2} \\ -\frac{T_{xxz}}{2} & T_{zxy} & -\frac{T_{zzx}}{2} \\ -\frac{T_{xxy}}{2} & -\frac{T_{yyx}}{2} & -T_{yzx} - T_{zxy} \end{bmatrix}$$

$$T_{xxy} = -2\mathbb{M}_{xz}^{(1)} + 2\mathbb{T}_{y}^{(2)}$$
$$T_{xxz} = 2\mathbb{M}_{xy}^{(1)} + 2\mathbb{T}_{z}^{(2)}$$
$$T_{yyx} = 2\mathbb{M}_{yz}^{(1)} + 2\mathbb{T}_{x}^{(2)}$$
$$T_{yyz} = -2\mathbb{M}_{xy}^{(1)} + 2\mathbb{T}_{z}^{(2)}$$
$$T_{zzx} = -2\mathbb{M}_{yz}^{(1)} + 2\mathbb{T}_{x}^{(2)}$$
$$T_{zzy} = 2\mathbb{M}_{xz}^{(1)} + 2\mathbb{T}_{y}^{(2)}$$
$$T_{yzx} = 3\mathbb{M}_{u}^{(1)} + \mathbb{M}_{v}^{(1)}$$
$$\frac{T_{yyz}}{2} = -\mathbb{M}_{xy}^{(1)} + \mathbb{T}_{z}^{(2)}$$
$$\frac{T_{zzy}}{2} = \mathbb{M}_{xz}^{(1)} + \mathbb{T}_{y}^{(2)}$$
$$\frac{T_{xxz}}{2} = \mathbb{M}_{xy}^{(1)} + \mathbb{T}_{z}^{(2)}$$
$$T_{zxy} = -3\mathbb{M}_{u}^{(1)} + \mathbb{M}_{v}^{(1)}$$
$$\frac{T_{zzx}}{2} = -\mathbb{M}_{yz}^{(1)} + \mathbb{T}_{x}^{(2)}$$
$$\frac{T_{xxy}}{2} = -\mathbb{M}_{xz}^{(1)} + \mathbb{T}_{y}^{(2)}$$
$$\frac{T_{yyx}}{2} = \mathbb{M}_{yz}^{(1)} + \mathbb{T}_{x}^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} T_{yzx} & T_{yzy} & T_{yzz} \\ T_{zxx} & T_{zxy} & T_{zxz} \\ T_{xyx} & T_{xyy} & T_{xyz} \end{bmatrix}$$

$$T_{yzx} = \mathbb{M}_0^{(1)} - \mathbb{M}_u^{(2)} + \mathbb{M}_v^{(2)}$$
$$T_{yzy} = \mathbb{M}_{xy}^{(2)} + \mathbb{T}_z^{(3)}$$
$$T_{yzz} = \mathbb{M}_{xz}^{(2)} - \mathbb{T}_y^{(3)}$$
$$T_{zxx} = \mathbb{M}_{xy}^{(2)} - \mathbb{T}_z^{(3)}$$
$$T_{zxy} = \mathbb{M}_0^{(1)} - \mathbb{M}_u^{(2)} - \mathbb{M}_v^{(2)}$$
$$T_{zxz} = \mathbb{M}_{yz}^{(2)} + \mathbb{T}_x^{(3)}$$
$$T_{xyx} = \mathbb{M}_{xz}^{(2)} + \mathbb{T}_y^{(3)}$$
$$T_{xyy} = \mathbb{M}_{yz}^{(2)} - \mathbb{T}_x^{(3)}$$
$$T_{xyz} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_u^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} T_{xxxx} & T_{xxyy} & T_{xxzz} & T_{xxyz} & T_{xxzx} & T_{xxxy} \\ T_{xxyy} & T_{yyyy} & T_{yyzz} & T_{yyyz} & T_{yyzx} & T_{yyxy} \\ T_{xxzz} & T_{yyzz} & T_{zzzz} & T_{zzyz} & T_{zzzx} & T_{zzxy} \\ T_{xxyz} & T_{yyyz} & T_{zzyz} & T_{yzyz} & T_{yzzx} & T_{yzxy} \\ T_{xxzx} & T_{yyzx} & T_{zzzx} & T_{yzzx} & T_{zxzx} & T_{zxxy} \\ T_{xxxy} & T_{yyxy} & T_{zzxy} & T_{yzxy} & T_{zxxy} & T_{xyxy} \end{bmatrix}$$

$$T_{xxxx} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} + 2\mathbb{T}_v^{(1)} + 4\mathbb{T}_v^{(2)}$$
$$T_{xxyy} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)}$$
$$T_{xxzz} = \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} + \mathbb{T}_v^{(1)}$$
$$T_{xxyz} = 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)}$$
$$T_{xxzx} = -\mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)}$$
$$T_{xxxy} = \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)}$$
$$T_{yyyy} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} - 2\mathbb{T}_v^{(1)} - 4\mathbb{T}_v^{(2)}$$
$$T_{yyzz} = \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} - \mathbb{T}_v^{(1)}$$
$$T_{yyyz} = \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)}$$
$$T_{yyzx} = 2\mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)}$$
$$T_{yyxy} = -\mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)}$$
$$T_{zzzz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} + 2\mathbb{T}_4^{(1)} + 4\mathbb{T}_u^{(1)} + 8\mathbb{T}_u^{(2)}$$
$$T_{zzyz} = -\mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)}$$
$$T_{zzzx} = \mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)}$$
$$T_{zzxy} = 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)}$$
$$T_{yzyz} = \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} - \mathbb{T}_v^{(2)}$$
$$T_{yzzx} = 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)}$$
$$T_{yzxy} = 2\mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(2)}$$
$$T_{zxzx} = \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} + \mathbb{T}_v^{(2)}$$
$$T_{zxxy} = 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(2)}$$
$$T_{xyxy} = \mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & T_{xxyy} & T_{xxzz} & T_{xxyz} & T_{xxzx} & T_{xxxy} \\ -T_{xxyy} & 0 & T_{yyzz} & T_{yyyz} & T_{yyzx} & T_{yyxy} \\ -T_{xxzz} & -T_{yyzz} & 0 & T_{zzyz} & T_{zzzx} & T_{zzxy} \\ -T_{xxyz} & -T_{yyyz} & -T_{zzyz} & 0 & T_{yzzx} & T_{yzxy} \\ -T_{xxzx} & -T_{yyzx} & -T_{zzzx} & -T_{yzzx} & 0 & T_{zxxy} \\ -T_{xxxy} & -T_{yyxy} & -T_{zzxy} & -T_{yzxy} & -T_{zxxy} & 0 \end{bmatrix}$$

$$T_{xxyy} = 4\mathbb{M}^{(1)}_{xyz} - 2\mathbb{T}^{(3)}_{v}$$
$$T_{xxzz} = -4\mathbb{M}^{(1)}_{xyz} + 3\mathbb{T}^{(3)}_{u} - \mathbb{T}^{(3)}_{v}$$
$$T_{xxyz} = -4\mathbb{M}^{\beta(1)}_{x} + \mathbb{T}^{(3)}_{yz}$$
$$T_{xxzx} = -2\mathbb{M}^{(1)}_{y} + 2\mathbb{M}^{\alpha(1)}_{y} + 2\mathbb{M}^{\beta(1)}_{y} + \mathbb{T}^{(3)}_{xz}$$
$$T_{xxxy} = 2\mathbb{M}^{(1)}_{z} - 2\mathbb{M}^{\alpha(1)}_{z} + 2\mathbb{M}^{\beta(1)}_{z} + \mathbb{T}^{(3)}_{xy}$$
$$T_{yyzz} = 4\mathbb{M}^{(1)}_{xyz} + 3\mathbb{T}^{(3)}_{u} + \mathbb{T}^{(3)}_{v}$$
$$T_{yyyz} = 2\mathbb{M}^{(1)}_{x} - 2\mathbb{M}^{\alpha(1)}_{x} + 2\mathbb{M}^{\beta(1)}_{x} + \mathbb{T}^{(3)}_{yz}$$
$$T_{yyzx} = -4\mathbb{M}^{\beta(1)}_{y} + \mathbb{T}^{(3)}_{xz}$$
$$T_{yyxy} = -2\mathbb{M}^{(1)}_{z} + 2\mathbb{M}^{\alpha(1)}_{z} + 2\mathbb{M}^{\beta(1)}_{z} + \mathbb{T}^{(3)}_{xy}$$
$$T_{zzyz} = -2\mathbb{M}^{(1)}_{x} + 2\mathbb{M}^{\alpha(1)}_{x} + 2\mathbb{M}^{\beta(1)}_{x} + \mathbb{T}^{(3)}_{yz}$$
$$T_{zzzx} = 2\mathbb{M}^{(1)}_{y} - 2\mathbb{M}^{\alpha(1)}_{y} + 2\mathbb{M}^{\beta(1)}_{y} + \mathbb{T}^{(3)}_{xz}$$
$$T_{zzxy} = -4\mathbb{M}^{\beta(1)}_{z} + \mathbb{T}^{(3)}_{xy}$$
$$T_{yzzx} = -\mathbb{M}^{(1)}_{z} - 4\mathbb{M}^{\alpha(1)}_{z}$$
$$T_{yzxy} = \mathbb{M}^{(1)}_{y} + 4\mathbb{M}^{\alpha(1)}_{y}$$
$$T_{zxxy} = -\mathbb{M}^{(1)}_{x} - 4\mathbb{M}^{\alpha(1)}_{x}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} T_{yzyz} & T_{yzzx} & T_{yzxy} \\ T_{yzzx} & T_{zxzx} & T_{zxxy} \\ T_{yzxy} & T_{zxxy} & T_{xyxy} \end{bmatrix}$$

$$T_{yzyz} = \mathbb{T}^{(3)}_{0} - 2\mathbb{T}^{(4)}_{u} + 2\mathbb{T}^{(4)}_{v}$$
$$T_{yzzx} = 2\mathbb{T}^{(4)}_{xy}$$
$$T_{yzxy} = 2\mathbb{T}^{(4)}_{xz}$$
$$T_{zxzx} = \mathbb{T}^{(3)}_{0} - 2\mathbb{T}^{(4)}_{u} - 2\mathbb{T}^{(4)}_{v}$$
$$T_{zxxy} = 2\mathbb{T}^{(4)}_{yz}$$
$$T_{xyxy} = \mathbb{T}^{(3)}_{0} + 4\mathbb{T}^{(4)}_{u}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & T_{yzzx} & T_{yzxy} \\ -T_{yzzx} & 0 & T_{zxxy} \\ -T_{yzxy} & -T_{zxxy} & 0 \end{bmatrix}$$

$$T_{yzzx} = -\mathbb{M}^{(2)}_{z}$$
$$T_{yzxy} = \mathbb{M}^{(2)}_{y}$$
$$T_{zxxy} = -\mathbb{M}^{(2)}_{x}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} T_{xxyz} & T_{xxzx} & T_{xxxy} \\ T_{yyyz} & T_{yyzx} & T_{yyxy} \\ T_{zzyz} & T_{zzzx} & T_{zzxy} \\ T_{yzyz} & T_{yzzx} & T_{yzxy} \\ T_{zxyz} & T_{zxzx} & T_{zxxy} \\ T_{xyyz} & T_{xyzx} & T_{xyxy} \end{bmatrix}$$

$$
\begin{aligned}
T_{xxyz} &= \mathbb{M}_{x}^{(3)} + 2\mathbb{M}_{x}^{(4)} + 2\mathbb{M}_{x}^{\alpha(2)} \\
T_{xxzx} &= \mathbb{M}_{y}^{(3)} - \mathbb{M}_{y}^{\alpha(2)} - \mathbb{M}_{y}^{\beta(2)} + 2\mathbb{T}_{xz}^{(5)} \\
T_{xxxy} &= \mathbb{M}_{z}^{(3)} - \mathbb{M}_{z}^{\alpha(2)} + \mathbb{M}_{z}^{\beta(2)} - 2\mathbb{T}_{xy}^{(5)} \\
T_{yyyz} &= \mathbb{M}_{x}^{(3)} - \mathbb{M}_{x}^{\alpha(2)} + \mathbb{M}_{x}^{\beta(2)} - 2\mathbb{T}_{yz}^{(5)} \\
T_{yyzx} &= \mathbb{M}_{y}^{(3)} + 2\mathbb{M}_{y}^{(4)} + 2\mathbb{M}_{y}^{\alpha(2)} \\
T_{yyxy} &= \mathbb{M}_{z}^{(3)} - \mathbb{M}_{z}^{\alpha(2)} - \mathbb{M}_{z}^{\beta(2)} + 2\mathbb{T}_{xy}^{(5)} \\
T_{zzyz} &= \mathbb{M}_{x}^{(3)} - \mathbb{M}_{x}^{\alpha(2)} - \mathbb{M}_{x}^{\beta(2)} + 2\mathbb{T}_{yz}^{(5)} \\
T_{zzzx} &= \mathbb{M}_{y}^{(3)} - \mathbb{M}_{y}^{\alpha(2)} + \mathbb{M}_{y}^{\beta(2)} - 2\mathbb{T}_{xz}^{(5)} \\
T_{zzxy} &= \mathbb{M}_{z}^{(3)} + 2\mathbb{M}_{z}^{(4)} + 2\mathbb{M}_{z}^{\alpha(2)} \\
T_{yzyz} &= \mathbb{M}_{xyz}^{(2)} - 3\mathbb{T}_{u}^{(5)} - \mathbb{T}_{v}^{(5)} \\
T_{yzzx} &= \mathbb{M}_{z}^{(4)} - \mathbb{M}_{z}^{\alpha(2)} - \mathbb{M}_{z}^{\beta(2)} - \mathbb{T}_{xy}^{(5)} \\
T_{yzxy} &= \mathbb{M}_{y}^{(4)} - \mathbb{M}_{y}^{\alpha(2)} + \mathbb{M}_{y}^{\beta(2)} + \mathbb{T}_{xz}^{(5)} \\
T_{zxyz} &= \mathbb{M}_{z}^{(4)} - \mathbb{M}_{z}^{\alpha(2)} + \mathbb{M}_{z}^{\beta(2)} + \mathbb{T}_{xy}^{(5)} \\
T_{zxzx} &= \mathbb{M}_{xyz}^{(2)} + 3\mathbb{T}_{u}^{(5)} - \mathbb{T}_{v}^{(5)} \\
T_{zxxy} &= \mathbb{M}_{x}^{(4)} - \mathbb{M}_{x}^{\alpha(2)} - \mathbb{M}_{x}^{\beta(2)} - \mathbb{T}_{yz}^{(5)} \\
T_{xyyz} &= \mathbb{M}_{y}^{(4)} - \mathbb{M}_{y}^{\alpha(2)} - \mathbb{M}_{y}^{\beta(2)} - \mathbb{T}_{xz}^{(5)} \\
T_{xyzx} &= \mathbb{M}_{x}^{(4)} - \mathbb{M}_{x}^{\alpha(2)} + \mathbb{M}_{x}^{\beta(2)} + \mathbb{T}_{yz}^{(5)} \\
T_{xyxy} &= \mathbb{M}_{xyz}^{(2)} + 2\mathbb{T}_{v}^{(5)}
\end{aligned}
$$

* Rank 4 tensor (as).

$$
\begin{bmatrix}
T_{yzxx} & T_{yzyy} & T_{yzzz} & T_{yzyz} & T_{yzzx} & T_{yzxy} \\
T_{zxxx} & T_{zxyy} & T_{zxzz} & T_{zxyz} & T_{zxzx} & T_{zxxy} \\
T_{xyxx} & T_{xyyy} & T_{xyzz} & T_{xyyz} & T_{xyzx} & T_{xyxy}
\end{bmatrix}
$$

$$
\begin{aligned}
T_{yzxx} &= \mathbb{M}_{x}^{(5)} + 2\mathbb{M}_{x}^{(6)} + 2\mathbb{M}_{x}^{\alpha(3)} \\
T_{yzyy} &= \mathbb{M}_{x}^{(5)} - \mathbb{M}_{x}^{\alpha(3)} + \mathbb{M}_{x}^{\beta(3)} - 2\mathbb{T}_{yz}^{(6)} \\
T_{yzzz} &= \mathbb{M}_{x}^{(5)} - \mathbb{M}_{x}^{\alpha(3)} - \mathbb{M}_{x}^{\beta(3)} + 2\mathbb{T}_{yz}^{(6)} \\
T_{yzyz} &= \mathbb{M}_{xyz}^{(3)} - 3\mathbb{T}_{u}^{(6)} - \mathbb{T}_{v}^{(6)} \\
T_{yzzx} &= \mathbb{M}_{z}^{(6)} - \mathbb{M}_{z}^{\alpha(3)} + \mathbb{M}_{z}^{\beta(3)} + \mathbb{T}_{xy}^{(6)} \\
T_{yzxy} &= \mathbb{M}_{y}^{(6)} - \mathbb{M}_{y}^{\alpha(3)} - \mathbb{M}_{y}^{\beta(3)} - \mathbb{T}_{xz}^{(6)} \\
T_{zxxx} &= \mathbb{M}_{y}^{(5)} - \mathbb{M}_{y}^{\alpha(3)} - \mathbb{M}_{y}^{\beta(3)} + 2\mathbb{T}_{xz}^{(6)} \\
T_{zxyy} &= \mathbb{M}_{y}^{(5)} + 2\mathbb{M}_{y}^{(6)} + 2\mathbb{M}_{y}^{\alpha(3)} \\
T_{zxzz} &= \mathbb{M}_{y}^{(5)} - \mathbb{M}_{y}^{\alpha(3)} + \mathbb{M}_{y}^{\beta(3)} - 2\mathbb{T}_{xz}^{(6)} \\
T_{zxyz} &= \mathbb{M}_{z}^{(6)} - \mathbb{M}_{z}^{\alpha(3)} - \mathbb{M}_{z}^{\beta(3)} - \mathbb{T}_{xy}^{(6)} \\
T_{zxzx} &= \mathbb{M}_{xyz}^{(3)} + 3\mathbb{T}_{u}^{(6)} - \mathbb{T}_{v}^{(6)} \\
T_{zxxy} &= \mathbb{M}_{x}^{(6)} - \mathbb{M}_{x}^{\alpha(3)} + \mathbb{M}_{x}^{\beta(3)} + \mathbb{T}_{yz}^{(6)} \\
T_{xyxx} &= \mathbb{M}_{z}^{(5)} - \mathbb{M}_{z}^{\alpha(3)} + \mathbb{M}_{z}^{\beta(3)} - 2\mathbb{T}_{xy}^{(6)} \\
T_{xyyy} &= \mathbb{M}_{z}^{(5)} - \mathbb{M}_{z}^{\alpha(3)} - \mathbb{M}_{z}^{\beta(3)} + 2\mathbb{T}_{xy}^{(6)} \\
T_{xyzz} &= \mathbb{M}_{z}^{(5)} + 2\mathbb{M}_{z}^{(6)} + 2\mathbb{M}_{z}^{\alpha(3)} \\
T_{xyyz} &= \mathbb{M}_{y}^{(6)} - \mathbb{M}_{y}^{\alpha(3)} + \mathbb{M}_{y}^{\beta(3)} + \mathbb{T}_{xz}^{(6)} \\
T_{xyzx} &= \mathbb{M}_{x}^{(6)} - \mathbb{M}_{x}^{\alpha(3)} - \mathbb{M}_{x}^{\beta(3)} - \mathbb{T}_{yz}^{(6)} \\
T_{xyxy} &= \mathbb{M}_{xyz}^{(3)} + 2\mathbb{T}_{v}^{(6)}
\end{aligned}
$$

* Rank 4 tensor (s).

$$\begin{bmatrix}
T_{xxxx} & T_{xxyy} & T_{xxzz} & T_{xxyz} & T_{xxzx} & T_{xxxy} & T_{xxzy} & T_{xxxz} & T_{xxyx} \\
T_{yyxx} & T_{yyyy} & T_{yyzz} & T_{yyyz} & T_{yyzx} & T_{yyxy} & T_{yyzy} & T_{yyxz} & T_{yyyx} \\
T_{zzxx} & T_{zzyy} & T_{zzzz} & T_{zzyz} & T_{zzzx} & T_{zzxy} & T_{zzzy} & T_{zzxz} & T_{zzyx} \\
T_{yzxx} & T_{yzyy} & T_{yzzz} & T_{yzyz} & T_{yzzx} & T_{yzxy} & T_{yzzy} & T_{yzxz} & T_{yzyx} \\
T_{zxxx} & T_{zxyy} & T_{zxzz} & T_{zxyz} & T_{zxzx} & T_{zxxy} & T_{zxzy} & T_{zxxz} & T_{zxyx} \\
T_{xyxx} & T_{xyyy} & T_{xyzz} & T_{xyyz} & T_{xyzx} & T_{xyxy} & T_{xyzy} & T_{xyxz} & T_{xyyx}
\end{bmatrix}$$

$$T_{xxxx} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} + 2\mathbb{T}_v^{(1)} + 4\mathbb{T}_v^{(2)}$$

$$T_{xxyy} = 4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 2\mathbb{T}_v^{(3)}$$

$$T_{xxzz} = -4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} + 3\mathbb{T}_u^{(3)} + \mathbb{T}_v^{(1)} - \mathbb{T}_v^{(3)}$$

$$T_{xxyz} = \mathbb{M}_x^{(3)} + 2\mathbb{M}_x^{(4)} + 2\mathbb{M}_x^{\alpha(2)} - 4\mathbb{M}_x^{\beta(1)} + 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + \mathbb{T}_{yz}^{(3)}$$

$$T_{xxzx} = -2\mathbb{M}_y^{(1)} + \mathbb{M}_y^{(3)} + 2\mathbb{M}_y^{\alpha(1)} - \mathbb{M}_y^{\alpha(2)} + 2\mathbb{M}_y^{\beta(1)} - \mathbb{M}_y^{\beta(2)} - \mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} + 2\mathbb{T}_{xz}^{(5)}$$

$$T_{xxxy} = 2\mathbb{M}_z^{(1)} + \mathbb{M}_z^{(3)} - 2\mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\alpha(2)} + 2\mathbb{M}_z^{\beta(1)} + \mathbb{M}_z^{\beta(2)} + \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} - 2\mathbb{T}_{xy}^{(5)}$$

$$T_{xxzy} = -\mathbb{M}_x^{(3)} - 2\mathbb{M}_x^{(4)} - 2\mathbb{M}_x^{\alpha(2)} - 4\mathbb{M}_x^{\beta(1)} + 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + \mathbb{T}_{yz}^{(3)}$$

$$T_{xxxz} = -2\mathbb{M}_y^{(1)} - \mathbb{M}_y^{(3)} + 2\mathbb{M}_y^{\alpha(1)} + \mathbb{M}_y^{\alpha(2)} + 2\mathbb{M}_y^{\beta(1)} + \mathbb{M}_y^{\beta(2)} - \mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} - 2\mathbb{T}_{xz}^{(5)}$$

$$T_{xxyx} = 2\mathbb{M}_z^{(1)} - \mathbb{M}_z^{(3)} - 2\mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\alpha(2)} + 2\mathbb{M}_z^{\beta(1)} - \mathbb{M}_z^{\beta(2)} + \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} + 2\mathbb{T}_{xy}^{(5)}$$

$$T_{yyxx} = -4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} + 2\mathbb{T}_v^{(3)}$$

$$T_{yyyy} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} - 2\mathbb{T}_v^{(1)} - 4\mathbb{T}_v^{(2)}$$

$$T_{yyzz} = 4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} + 3\mathbb{T}_u^{(3)} - \mathbb{T}_v^{(1)} + \mathbb{T}_v^{(3)}$$

$$T_{yyyz} = 2\mathbb{M}_x^{(1)} + \mathbb{M}_x^{(3)} - 2\mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\alpha(2)} + 2\mathbb{M}_x^{\beta(1)} + \mathbb{M}_x^{\beta(2)} + \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)}$$

$$T_{yyzx} = \mathbb{M}_y^{(3)} + 2\mathbb{M}_y^{(4)} + 2\mathbb{M}_y^{\alpha(2)} - 4\mathbb{M}_y^{\beta(1)} + 2\mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + \mathbb{T}_{xz}^{(3)}$$

$$T_{yyxy} = -2\mathbb{M}_z^{(1)} + \mathbb{M}_z^{(3)} + 2\mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\alpha(2)} + 2\mathbb{M}_z^{\beta(1)} - \mathbb{M}_z^{\beta(2)} - \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} + 2\mathbb{T}_{xy}^{(5)}$$

$$T_{yyzy} = 2\mathbb{M}_x^{(1)} - \mathbb{M}_x^{(3)} - 2\mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\alpha(2)} + 2\mathbb{M}_x^{\beta(1)} - \mathbb{M}_x^{\beta(2)} + \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} + 2\mathbb{T}_{yz}^{(5)}$$

$$T_{yyzx} = -\mathbb{M}_y^{(3)} - 2\mathbb{M}_y^{(4)} - 2\mathbb{M}_y^{\alpha(2)} - 4\mathbb{M}_y^{\beta(1)} + 2\mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + \mathbb{T}_{xz}^{(3)}$$

$$T_{yyyx} = -2\mathbb{M}_z^{(1)} - \mathbb{M}_z^{(3)} + 2\mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\alpha(2)} + 2\mathbb{M}_z^{\beta(1)} + \mathbb{M}_z^{\beta(2)} - \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} - 2\mathbb{T}_{xy}^{(5)}$$

$$T_{zzxx} = 4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} - 3\mathbb{T}_u^{(3)} + \mathbb{T}_v^{(1)} + \mathbb{T}_v^{(3)}$$

$$T_{zzyy} = -4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} - 3\mathbb{T}_u^{(3)} - \mathbb{T}_v^{(1)} - \mathbb{T}_v^{(3)}$$

$$T_{zzzz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} + 2\mathbb{T}_4^{(1)} + 4\mathbb{T}_u^{(1)} + 8\mathbb{T}_u^{(2)}$$

$$T_{zzyz} = -2\mathbb{M}_x^{(1)} + \mathbb{M}_x^{(3)} + 2\mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\alpha(2)} + 2\mathbb{M}_x^{\beta(1)} - \mathbb{M}_x^{\beta(2)} - \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} + 2\mathbb{T}_{yz}^{(5)}$$

$$T_{zzzx} = 2\mathbb{M}_y^{(1)} + \mathbb{M}_y^{(3)} - 2\mathbb{M}_y^{\alpha(1)} - \mathbb{M}_y^{\alpha(2)} + 2\mathbb{M}_y^{\beta(1)} + \mathbb{M}_y^{\beta(2)} + \mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} - 2\mathbb{T}_{xz}^{(5)}$$

$$T_{zzxy} = \mathbb{M}_z^{(3)} + 2\mathbb{M}_z^{(4)} + 2\mathbb{M}_z^{\alpha(2)} - 4\mathbb{M}_z^{\beta(1)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + \mathbb{T}_{xy}^{(3)}$$

$$T_{zzzy} = -2\mathbb{M}_x^{(1)} - \mathbb{M}_x^{(3)} + 2\mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\alpha(2)} + 2\mathbb{M}_x^{\beta(1)} + \mathbb{M}_x^{\beta(2)} - \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)}$$

$$T_{zzxz} = 2\mathbb{M}_y^{(1)} - \mathbb{M}_y^{(3)} - 2\mathbb{M}_y^{\alpha(1)} + \mathbb{M}_y^{\alpha(2)} + 2\mathbb{M}_y^{\beta(1)} - \mathbb{M}_y^{\beta(2)} + \mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} + 2\mathbb{T}_{xz}^{(5)}$$

$$T_{zzyx} = -\mathbb{M}_z^{(3)} - 2\mathbb{M}_z^{(4)} - 2\mathbb{M}_z^{\alpha(2)} - 4\mathbb{M}_z^{\beta(1)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + \mathbb{T}_{xy}^{(3)}$$

$$T_{yzxx} = 4\mathbb{M}_x^{\beta(1)} + 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} - \mathbb{T}_{yz}^{(3)}$$

$$T_{yzyy} = -2\mathbb{M}_x^{(1)} + 2\mathbb{M}_x^{\alpha(1)} - 2\mathbb{M}_x^{\beta(1)} + \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(3)}$$

$$T_{yzzz} = 2\mathbb{M}_x^{(1)} - 2\mathbb{M}_x^{\alpha(1)} - 2\mathbb{M}_x^{\beta(1)} - \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(3)}$$

$$T_{yzyz} = \mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} - 3\mathbb{T}_u^{(5)} - \mathbb{T}_v^{(2)} - \mathbb{T}_v^{(5)}$$

$$T_{yzzx} = -\mathbb{M}_z^{(1)} + \mathbb{M}_z^{(4)} - 4\mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\alpha(2)} - \mathbb{M}_z^{\beta(2)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(5)}$$

$$T_{yzxy} = \mathbb{M}_y^{(1)} + \mathbb{M}_y^{(4)} + 4\mathbb{M}_y^{\alpha(1)} - \mathbb{M}_y^{\alpha(2)} + \mathbb{M}_y^{\beta(2)} + 2\mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(5)}$$

$$T_{yzzy} = -\mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} + 3\mathbb{T}_u^{(5)} - \mathbb{T}_v^{(2)} + \mathbb{T}_v^{(5)}$$

$$T_{yzxz} = -\mathbb{M}_z^{(1)} - \mathbb{M}_z^{(4)} - 4\mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\alpha(2)} + \mathbb{M}_z^{\beta(2)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(5)}$$

$$T_{yzyx} = \mathbb{M}_y^{(1)} - \mathbb{M}_y^{(4)} + 4\mathbb{M}_y^{\alpha(1)} + \mathbb{M}_y^{\alpha(2)} - \mathbb{M}_y^{\beta(2)} + 2\mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(5)}$$

$$T_{zxxx} = 2\mathbb{M}_y^{(1)} - 2\mathbb{M}_y^{\alpha(1)} - 2\mathbb{M}_y^{\beta(1)} - \mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(3)}$$

$$T_{zxyy} = 4\mathbb{M}_y^{\beta(1)} + 2\mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} - \mathbb{T}_{xz}^{(3)}$$

$$T_{zxzz} = -2\mathbb{M}_y^{(1)} + 2\mathbb{M}_y^{\alpha(1)} - 2\mathbb{M}_y^{\beta(1)} + \mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(3)}$$

$$T_{zxyz} = \mathbb{M}_z^{(1)} + \mathbb{M}_z^{(4)} + 4\mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\alpha(2)} + \mathbb{M}_z^{\beta(2)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(5)}$$

$$T_{zxzx} = \mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} + 3\mathbb{T}_u^{(5)} + \mathbb{T}_v^{(2)} - \mathbb{T}_v^{(5)}$$

$$T_{zxxy} = -\mathbb{M}_x^{(1)} + \mathbb{M}_x^{(4)} - 4\mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\alpha(2)} - \mathbb{M}_x^{\beta(2)} + 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(5)}$$

$$T_{zxzy} = \mathbb{M}_z^{(1)} - \mathbb{M}_z^{(4)} + 4\mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\alpha(2)} - \mathbb{M}_z^{\beta(2)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(5)}$$

$$T_{zxxz} = -\mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} - 3\mathbb{T}_u^{(5)} + \mathbb{T}_v^{(2)} + \mathbb{T}_v^{(5)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} T_{yzxx} & T_{yzyy} & T_{yzzz} & T_{yzyz} & T_{yzzx} & T_{yzxy} & T_{yzzy} & T_{yzxz} & T_{yzyx} \\ T_{zxxx} & T_{zxyy} & T_{zxzz} & T_{zxyz} & T_{zxzx} & T_{zxxy} & T_{zxzy} & T_{zxxz} & T_{zxyx} \\ T_{xyxx} & T_{xyyy} & T_{xyzz} & T_{xyyz} & T_{xyzx} & T_{xyxy} & T_{xyzy} & T_{xyxz} & T_{xyyx} \end{bmatrix}$$

$$\begin{aligned}
T_{yzxx} &= \mathbb{M}_x^{(5)} + 2\mathbb{M}_x^{(6)} + 2\mathbb{M}_x^{\alpha(3)} \\
T_{yzyy} &= \mathbb{M}_x^{(5)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(6)} \\
T_{yzzz} &= \mathbb{M}_x^{(5)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(6)} \\
T_{yzyz} &= \mathbb{M}_{xyz}^{(3)} + \mathbb{T}_0^{(3)} - 2\mathbb{T}_u^{(4)} - 3\mathbb{T}_u^{(6)} + 2\mathbb{T}_v^{(4)} - \mathbb{T}_v^{(6)} \\
T_{yzzx} &= -\mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} + 2\mathbb{T}_{xy}^{(4)} + \mathbb{T}_{xy}^{(6)} \\
T_{yzxy} &= \mathbb{M}_y^{(2)} + \mathbb{M}_y^{(6)} - \mathbb{M}_y^{\alpha(3)} - \mathbb{M}_y^{\beta(3)} + 2\mathbb{T}_{xz}^{(4)} - \mathbb{T}_{xz}^{(6)} \\
T_{yzzy} &= \mathbb{M}_{xyz}^{(3)} - \mathbb{T}_0^{(3)} + 2\mathbb{T}_u^{(4)} - 3\mathbb{T}_u^{(6)} - 2\mathbb{T}_v^{(4)} - \mathbb{T}_v^{(6)} \\
T_{yzxz} &= \mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} - 2\mathbb{T}_{xy}^{(4)} + \mathbb{T}_{xy}^{(6)} \\
T_{yzyx} &= -\mathbb{M}_y^{(2)} + \mathbb{M}_y^{(6)} - \mathbb{M}_y^{\alpha(3)} - \mathbb{M}_y^{\beta(3)} - 2\mathbb{T}_{xz}^{(4)} - \mathbb{T}_{xz}^{(6)} \\
T_{zxxx} &= \mathbb{M}_y^{(5)} - \mathbb{M}_y^{\alpha(3)} - \mathbb{M}_y^{\beta(3)} + 2\mathbb{T}_{xz}^{(6)} \\
T_{zxyy} &= \mathbb{M}_y^{(5)} + 2\mathbb{M}_y^{(6)} + 2\mathbb{M}_y^{\alpha(3)} \\
T_{zxzz} &= \mathbb{M}_y^{(5)} - \mathbb{M}_y^{\alpha(3)} + \mathbb{M}_y^{\beta(3)} - 2\mathbb{T}_{xz}^{(6)} \\
T_{zxyz} &= \mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} + 2\mathbb{T}_{xy}^{(4)} - \mathbb{T}_{xy}^{(6)} \\
T_{zxzx} &= \mathbb{M}_{xyz}^{(3)} + \mathbb{T}_0^{(3)} - 2\mathbb{T}_u^{(4)} + 3\mathbb{T}_u^{(6)} - 2\mathbb{T}_v^{(4)} - \mathbb{T}_v^{(6)} \\
T_{zxxy} &= -\mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(4)} + \mathbb{T}_{yz}^{(6)} \\
T_{zxzy} &= -\mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} - 2\mathbb{T}_{xy}^{(4)} - \mathbb{T}_{xy}^{(6)} \\
T_{zxxz} &= \mathbb{M}_{xyz}^{(3)} - \mathbb{T}_0^{(3)} + 2\mathbb{T}_u^{(4)} + 3\mathbb{T}_u^{(6)} + 2\mathbb{T}_v^{(4)} - \mathbb{T}_v^{(6)} \\
T_{zxyx} &= \mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(4)} + \mathbb{T}_{yz}^{(6)} \\
T_{xyxx} &= \mathbb{M}_z^{(5)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} - 2\mathbb{T}_{xy}^{(6)} \\
T_{xyyy} &= \mathbb{M}_z^{(5)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} + 2\mathbb{T}_{xy}^{(6)} \\
T_{xyzz} &= \mathbb{M}_z^{(5)} + 2\mathbb{M}_z^{(6)} + 2\mathbb{M}_z^{\alpha(3)} \\
T_{xyyz} &= -\mathbb{M}_y^{(2)} + \mathbb{M}_y^{(6)} - \mathbb{M}_y^{\alpha(3)} + \mathbb{M}_y^{\beta(3)} + 2\mathbb{T}_{xz}^{(4)} + \mathbb{T}_{xz}^{(6)} \\
T_{xyzx} &= \mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(4)} - \mathbb{T}_{yz}^{(6)} \\
T_{xyxy} &= \mathbb{M}_{xyz}^{(3)} + \mathbb{T}_0^{(3)} + 4\mathbb{T}_u^{(4)} + 2\mathbb{T}_v^{(6)} \\
T_{xyzy} &= \mathbb{M}_y^{(2)} + \mathbb{M}_y^{(6)} - \mathbb{M}_y^{\alpha(3)} + \mathbb{M}_y^{\beta(3)} - 2\mathbb{T}_{xz}^{(4)} + \mathbb{T}_{xz}^{(6)} \\
T_{xyxz} &= -\mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(4)} - \mathbb{T}_{yz}^{(6)} \\
T_{xyyx} &= \mathbb{M}_{xyz}^{(3)} - \mathbb{T}_0^{(3)} - 4\mathbb{T}_u^{(4)} + 2\mathbb{T}_v^{(6)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} T_{xxxx} & T_{xxxy} & T_{xxxz} \\ T_{yyyx} & T_{yyyy} & T_{yyyz} \\ T_{zzzx} & T_{zzzy} & T_{zzzz} \\ T_{yyzx} & T_{yyzy} & T_{yyzz} \\ T_{zzxx} & T_{zzxy} & T_{zzxz} \\ T_{xxyx} & T_{xxyy} & T_{xxyz} \\ T_{yzzx} & T_{yzzy} & T_{yzzz} \\ T_{zxxx} & T_{zxxy} & T_{zxxz} \\ T_{xxyy} + T_{yyzz} - T_{yzzy} - T_{zxxz} + T_{zzxx} & T_{xyyy} & T_{xyyz} \\ -\frac{T_{yyyz}}{2} + \frac{T_{yyzy}}{2} + T_{zxxy} & T_{xyyz} + \frac{T_{zzzz}}{2} - \frac{T_{zzzx}}{2} & \frac{T_{xxxy}}{2} - \frac{T_{xxyx}}{2} + T_{xyyy} - T_{yyyx} + T_{zzxy} \end{bmatrix}$$

$$T_{xxxx} = 3\mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} + 2\mathbb{T}_v^{(1)} + 4\mathbb{T}_v^{(2)}$$
$$T_{xxxy} = \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} - 2\mathbb{T}_{xy}^{(5)}$$
$$T_{xxxz} = -\mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} - 2\mathbb{T}_{xz}^{(5)}$$
$$T_{yyyx} = -\mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} - 2\mathbb{T}_{xy}^{(5)}$$
$$T_{yyyy} = 3\mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} - 2\mathbb{T}_v^{(1)} - 4\mathbb{T}_v^{(2)}$$
$$T_{yyyz} = \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)}$$
$$T_{zzzx} = \mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} - 2\mathbb{T}_{xz}^{(5)}$$
$$T_{zzzy} = -\mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)}$$
$$T_{zzzz} = 3\mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} + 2\mathbb{T}_4^{(1)} + 4\mathbb{T}_u^{(1)} + 8\mathbb{T}_u^{(2)}$$
$$T_{yyzx} = 2\mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + \mathbb{T}_{xz}^{(3)}$$
$$T_{yyzy} = \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} + 2\mathbb{T}_{yz}^{(5)}$$
$$T_{yyzz} = \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} + 3\mathbb{T}_u^{(3)} - \mathbb{T}_v^{(1)} + \mathbb{T}_v^{(3)}$$
$$T_{zzxx} = \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} - 3\mathbb{T}_u^{(3)} + \mathbb{T}_v^{(1)} + \mathbb{T}_v^{(3)}$$
$$T_{zzxy} = 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + \mathbb{T}_{xy}^{(3)}$$
$$T_{zzxz} = \mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} + 2\mathbb{T}_{xz}^{(5)}$$
$$T_{xxyx} = \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} + 2\mathbb{T}_{xy}^{(5)}$$
$$T_{xxyy} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 2\mathbb{T}_v^{(3)}$$
$$T_{xxyz} = 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + \mathbb{T}_{yz}^{(3)}$$
$$T_{yzzx} = 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(5)}$$
$$T_{yzzy} = \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} + 3\mathbb{T}_u^{(5)} - \mathbb{T}_v^{(2)} + \mathbb{T}_v^{(5)}$$
$$T_{yzzz} = -\mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(3)}$$
$$T_{zxxx} = -\mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(3)}$$
$$T_{zxxy} = 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(5)}$$
$$T_{zxxz} = \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} - 3\mathbb{T}_u^{(5)} + \mathbb{T}_v^{(2)} + \mathbb{T}_v^{(5)}$$
$$T_{xyyy} = -\mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(3)}$$
$$T_{xyyz} = 2\mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(5)}$$